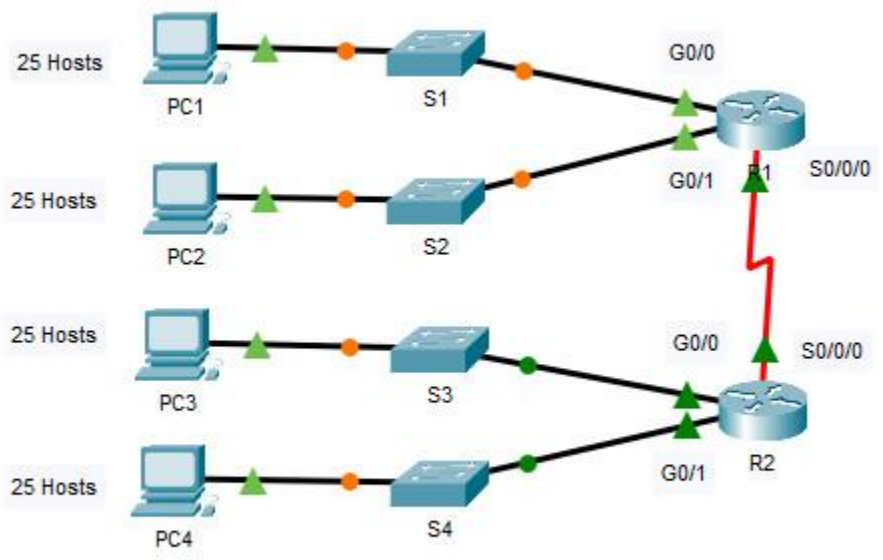
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Packet Tracer - Subnetting Scenario (Instructor Version)

Topology



11.7.5 Packet Tracer - Subnetting Scenario

Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	G0/0	192.168.100.1	255.255.255.224	N/A
	G0/1	192.168.100.33	255.255.255.224	N/A
	S0/0/0	192.168.100.129	255.255.255.224	N/A
R2	G0/0	192.168.100.65	255.255.255.224	N/A
	G0/1	192.168.100.97	255.255.255.224	N/A
	S0/0/0	192.168.100.158	255.255.255.224	N/A

Device	Interface	IP Address	Subnet Mask	Default Gateway
S1	VLAN 1	192.168.100.2	255.255.255.224	192.168.100.1
S2	VLAN 1	192.168.100.34	255.255.255.224	192.168.100.33
S3	VLAN 1	192.168.100.66	255.255.255.224	192.168.100.65
S4	VLAN 1	192.168.100.98	255.255.255.224	192.168.100.97
PC1	NIC	192.168.100.30	255.255.255.224	192.168.100.1
PC2	NIC	192.168.100.62	255.255.255.224	192.168.100.33
PC3	NIC	192.168.100.94	255.255.255.224	192.168.100.65
PC4	NIC	192.168.100.126	255.255.255.224	192.168.100.97

Objectives

- Part 1: Design an IP Addressing Scheme
- Part 2: Assign IP Addresses to Network Devices and Verify Connectivity

Scenario

In this activity, you are given the network address of 192.168.100.0/24 to subnet and provide the IP addressing for the Packet Tracer network. Each LAN in the network requires at least 25 addresses for end devices, the switch and the router. The connection between R1 to R2 will require an IP address for each end of the link.

Instructions

Part 1: Design an IP Addressing Scheme

Step 1: Subnet the 192.168.100.0/24 network into the appropriate number of subnets.

Questions:

a. Based on the topology, how many subnets are needed?

5. Four for the LANs, and one for the link between the routers.

b. How many bits must be borrowed to support the number of subnets in the topology table?

3

c. How many subnets does this create?

8

d. How many usable hosts does this create per subnet?

30

Note: If your answer is less than the 25 hosts required, then you borrowed too many bits.

e. Calculate the binary value for the first five subnets. The first two subnets have been done for you.

Subnet	Network Address	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
0	192.168.100.	0	0	0	0	0	0	0	0
1	192.168.100.	0	0	1	0	0	0	0	0
2	192.168.100.	0	1	0	0	0	0	0	0
3	192.168.100.	0	1	1	0	0	0	0	0
4	192.168.100.	1	0	0	0	0	0	0	0

f. Calculate the binary and decimal value of the new subnet mask.

First Octet	Second Octet	Third Octet	Mask Bit 7	Mask Bit 6	Mask Bit 5	Mask Bit 4	Mask Bit 3	Mask Bit 2	
11111111	11111111	11111111	1	1	1	0	0	0	
First Decimal Octet	**Second Decimal Octet**	**Third Decimal Octet**	**Fourth Decimal Octet**						

First Octet	Second Octet	Third Octet	Mask Bit 7	Mask Bit 6	Mask Bit 5	Mask Bit 4	Mask Bit 3	Mask Bit 2
255.	255.	255.	224					

g. Fill in the **Subnet Table**, listing the decimal value of all available subnets, the first and last usable host address, and the broadcast address. Repeat until all addresses are listed.

Note: You may not need to use all rows.

Subnet Table

Subnet Number	Subnet Address	First Usable Host Address	Last Usable Host Address	Broadcast Address
0	192.168.100.0	192.168.100.1	192.168.100.30	192.168.100.31
1	192.168.100.32	192.168.100.33	192.168.100.62	192.168.100.63
2	192.168.100.64	192.168.100.65	192.168.100.94	192.168.100.95
3	192.168.100.96	192.168.100.97	192.168.100.126	192.168.100.127
4	192.168.100.128	192.168.100.129	192.168.100.158	192.168.100.159
5	192.168.100.160	192.168.100.161	192.168.100.190	192.168.100.191
6	192.168.100.192	192.168.100.193	192.168.100.222	192.168.100.223
7	192.168.100.224	192.168.100.225	192.168.100.254	192.168.100.255
8				
9				
10				

💡 **Pro Tip: Struggling with subnetting?** Try our **IPv4 Subnet Calculator (FLSM)** to quickly check your results!

Step 2: Assign the subnets to the network shown in the topology.

a. Assign Subnet 0 to the LAN connected to the GigabitEthernet 0/0 interface of R1: 192.168.100.0/27

b. Assign Subnet 1 to the LAN connected to the GigabitEthernet 0/1 interface of R1: 192.168.100.32/27

c. Assign Subnet 2 to the LAN connected to the GigabitEthernet 0/0 interface of R2: 192.168.100.64/27

d. Assign Subnet 3 to the LAN connected to the GigabitEthernet 0/1 interface of R2: 192.168.100.96/27

e. Assign Subnet 4 to the WAN link between R1 to R2: 192.168.100.128/27

Step 3: Document the addressing scheme.

Fill in the **Addressing Table** using the following guidelines:
a. Assign the first usable IP addresses in each subnet to R1 for the two LAN links and the WAN link.

b. Assign the first usable IP addresses in each subnet to R2 for the LAN links. Assign the last usable IP address for the WAN link.

c. Assign the second usable IP address in the attached subnets to the switches.

d. Assign the last usable IP addresses to the PCs in each subnet.

Part 2: Assign IP Addresses to Network Devices and Verify Connectivity

Most of the IP addressing is already configured on this network. Implement the following steps to complete the addressing configuration. EIGRP dynamic routing is already configured between R1 and R2.

Step 1: Configure R1 LAN interfaces.

a. Configure both LAN interfaces with the addresses from the Addressing Table.

R1(config)# interface gigabitethernet 0/0

R1(config-if)# ip address 192.168.100.1 255.255.255.224

R1(config-if)# interface gigabitethernet 0/1

R1(config-if)# ip address 192.168.100.33 255.255.255.224

b. Configure the interfaces so that the hosts on the LANs have connectivity to the default gateway.

R1(config)# interface gigabitethernet 0/0

R1(config-if)# no shutdown

R1(config-if)# interface gigabitethernet 0/1

R1(config-if)# no shutdown

Step 2: Configure IP addressing on S3.

a. Configure the switch VLAN1 interface with addressing.

S3(config)# interface vlan 1

S3(config-if)# ip address 192.168.100.66 255.255.255.224

S3(config-if)# no shutdown

b. Configure the switch with the default gateway address.

S3(config)# ip default-gateway 192.168.100.65

Step 3: Configure PC4.

Configure PC4 with host and default gateway addresses.

Step 4: Verify connectivity.

You can only verify connectivity from R1, S3, and PC4. However, you should be able to ping every IP address listed in the **Addressing Table**.

R1# ping 192.168.100.126

S3# ping 192.168.100.1

C:\> ping 192.168.100.30

Device Configs - Final

ROUTER R1

! ==

!--- 11.7.5 Packet Tracer - Subnetting Scenario

!--- ANSWER SCRIPT FOR ROUTER R1

```
!--- Usage: copy this whole file and paste it into the R1 terminal (start at the R1> prompt).
Every line beginning with "!" is a comment; IOS ignores it, so pasting a comment by
accident is harmless.
! ============================================================


enable
configure terminal


! ------------------------------------------------------------
!--- Part 2: Assign IP Addresses to Network Devices and Verify Connectivity.
! ------------------------------------------------------------


!--- Step 1a/1b: G0/0 - first usable address of Subnet 0 (192.168.100.0/27); activate it so the
LAN hosts reach their default gateway.
interface gigabitethernet 0/0
 ip address 192.168.100.1 255.255.255.224
 no shutdown
 exit


!--- Step 1a/1b: G0/1 - first usable address of Subnet 1 (192.168.100.32/27); activate it.
interface gigabitethernet 0/1
 ip address 192.168.100.33 255.255.255.224
 no shutdown
 exit


end
```

! --

!--- Save the configuration to NVRAM. (Press Enter when prompted for the destination filename.)

! --

running-config startup-config

! ==

!--- Verification:

!--- show ip interface brief -> G0/0, G0/1 and S0/0/0 up/up with their addresses

!--- show ip route -> 3 connected (C) /27 subnets and 2 EIGRP (D) routes (192.168.100.64/27, 192.168.100.96/27)

!--- ping 192.168.100.66 -> reaches S3 (after S3 is configured)

!--- ping 192.168.100.126 -> reaches PC4 (after PC4 is configured)

!--- Note: S0/0/0 (192.168.100.129/27) and EIGRP are already configured on R1 and R2.

! ==

SWITCH S3

! ==

!--- 11.7.5 Packet Tracer - Subnetting Scenario

!--- ANSWER SCRIPT FOR SWITCH S3

!--- Usage: copy this whole file and paste it into the S3 terminal (start at the S3> prompt). Every line beginning with "!" is a comment; IOS ignores it, so pasting a comment by accident is harmless.

! ==

enable

configure terminal

```
! ------------------------------------------------------------

!--- Part 2: Assign IP Addresses to Network Devices and Verify Connectivity.

! ------------------------------------------------------------


!--- Step 2a: VLAN 1 - second usable address of Subnet 2 (192.168.100.64/27); activate it.

interface vlan 1

 ip address 192.168.100.66 255.255.255.224

 no shutdown

 exit


!--- Step 2b: default gateway - R2 G0/0.

ip default-gateway 192.168.100.65


end


! ------------------------------------------------------------

!--- Save the configuration to NVRAM. (Press Enter when prompted for the destination filename.)

! ------------------------------------------------------------

copy running-config startup-config



! ============================================================

!--- Verification:

!---   show ip interface brief  -> Vlan1 up/up with 192.168.100.66
```

!--- ping 192.168.100.65 -> reaches the default gateway (R2 G0/0)

!--- ping 192.168.100.1 -> reaches R1 G0/0 across the WAN link

! ==

PC4

PC4PC

IP Address192.168.100.126

Subnet Mask255.255.255.224

Default Gateway192.168.100.97

Command Prompt ipconfig 192.168.100.126 255.255.255.224 192.168.100.97

Cisco Packet Tracer — PC4 Command Prompt (IP Config & Ping Verification)

```
C:\>ipconfig 192.168.100.126 255.255.255.224 192.168.100.97
IP Address.....................: 192.168.100.126
Subnet Mask....................: 255.255.255.224
Default Gateway................: 192.168.100.97

C:\>ping 192.168.100.30
Pinging 192.168.100.30 with 32 bytes of data:
Reply from 192.168.100.30: bytes=32 time=1ms TTL=126
Reply from 192.168.100.30: bytes=32 time<1ms TTL=126
Reply from 192.168.100.30: bytes=32 time<1ms TTL=126
Reply from 192.168.100.30: bytes=32 time<1ms TTL=126
Ping statistics for 192.168.100.30: Sent = 4, Received = 4, Lost = 0 (0% loss)
```